

Тригонометрические и гиперболические замены: дополнение

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Theoretical minimum.

Def. A substitution $x = \varphi(\theta)$ *conjugates* a recurrence $x_{n+1} = F(x_n)$ to the map $\theta \mapsto k\theta$ if $F(\varphi(\theta)) = \varphi(k\theta)$. When this happens, $x_n = \varphi(k^n\theta_0)$ and the whole sequence is written in closed form at once.

Theorem 1 (the table). Four substitutions cover almost everything:

- a) $x = 2 \cos \theta$ for $|x| \leq 2$ and $x = 2 \operatorname{ch} t$ for $|x| \geq 2$ conjugate $x \mapsto x^2 - 2$ to doubling, since $(2 \cos \theta)^2 - 2 = 2 \cos 2\theta$;
- b) $x = \cos \theta$ conjugates $x \mapsto 2x^2 - 1$ to doubling and $x \mapsto \sqrt{\frac{1+x}{2}}$ to halving;
- c) $x = \operatorname{tg} \theta$ conjugates $x \mapsto \frac{2x}{1-x^2}$ to doubling and $x \mapsto \frac{x+a}{1-ax}$ to the shift $\theta \mapsto \theta + \operatorname{arctg} a$;
- d) $x = \sqrt{a} \operatorname{cth} t$ conjugates the Newton step $x \mapsto \frac{1}{2}(x + \frac{a}{x})$ to doubling, because $\operatorname{cth} 2t = \frac{1}{2}(\operatorname{cth} t + \frac{1}{\operatorname{cth} t})$.

Theorem 2 (Chebyshev behind the table). $T_n(\cos \theta) = \cos n\theta$ and $T_n(\operatorname{ch} t) = \operatorname{ch} nt$, so $T_m \circ T_n = T_{mn}$. Equivalently, $x = z + \frac{1}{z}$ conjugates $x \mapsto x^2 - 2$ to $z \mapsto z^2$: on the unit circle this is the cosine, outside it the hyperbolic cosine. All of case a) is this one observation.

Fact 1 (as распознать). Look for the substitution when the recurrence preserves an interval and looks like an addition theorem: a square minus a constant, a ratio $\frac{u+v}{1-uv}$, a half of a sum with a reciprocal, or a nested radical $\sqrt{c+x}$. A second signal is that the answer is expected in closed form for

every n : no other elementary device produces closed forms for a nonlinear recurrence.

Приём встречается и в других листках: Trig-3 (задачи 3 и 5 и домашние 6 и 9), Limits-2 (задачи 1 и 5). Здесь он собран как метод.

Class problems

1. Prove all four conjugations of Theorem 1 and Theorem 2. Deduce that the recurrence $a_{n+1} = a_n^2 - 2$ with $a_1 = t + \frac{1}{t}$ has the closed form $a_n = t^{2^{n-1}} + t^{-2^{n-1}}$, and say which of t real or t on the unit circle corresponds to $|a_1| > 2$ and to $|a_1| \leq 2$. The qualitative picture of this sequence was obtained in Trig-3 by hand; here it becomes a formula.
2. (IMO 1976) Let $P_1(x) = x^2 - 2$ and $P_j(x) = P_1(P_{j-1}(x))$ for $j \geq 2$. Prove that for every n the roots of the equation $P_n(x) = x$ are real and pairwise distinct.
3. (IMO shortlist) Let a be a positive integer and let $a_0 = 0$, $a_{n+1} = (a_n + 1)a + (a + 1)a_n + 2\sqrt{a(a + 1)a_n(a_n + 1)}$. Prove that every a_n is a positive integer. *Hint*: the right-hand side is the addition formula for sh^2 .

Homework

1. (Engel) Find a closed form for the number $a_n = \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}$ with n radicals, and compute $\lim_{n \rightarrow \infty} 4^n(2 - a_n)$.
2. (Engel) Solve the system $x^2 + y^2 = 1$, $4x^3 - 3x = \sqrt{\frac{x+1}{2}}$.
3. Prove that $x_{n+1} = \sqrt{\frac{1+x_n}{2}}$ with $x_0 = \cos \theta$ gives $x_n = \cos \frac{\theta}{2^n}$, and deduce Viète's formula $\prod_{k \geq 1} \cos \frac{\theta}{2^k} = \frac{\sin \theta}{\theta}$ together with its classical special case at $\theta = \frac{\pi}{2}$.

4. (IMO shortlist) Let $f: (0, +\infty) \rightarrow \mathbb{R}$ satisfy $f(x) = f\left(\frac{1}{x}\right)$ for all $x > 0$. Prove that there is $u: [2, +\infty) \rightarrow \mathbb{R}$ with $f(x) = u\left(x + \frac{1}{x}\right)$, and prove the rational version: a rational f with $f(x) = f\left(\frac{1}{x}\right)$ is a rational function of $x + \frac{1}{x}$.
5. Prove that the sequence $x_{n+1} = \frac{x_n + \sqrt{3}}{1 - \sqrt{3}x_n}$ is periodic for every starting value where it is defined, and find the period; do the same for $x_{n+1} = \frac{2x_n}{1 - x_n^2}$ with $x_0 = \operatorname{tg} \frac{\pi}{2^k}$.
6. (Engel) For the “divide and average” iteration $x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right)$ with $x_0 > 0$ the convergence to $\sqrt{a}$ was proved in Limits – 2; prove here the exact closed form $\frac{x_n - \sqrt{a}}{x_n + \sqrt{a}} = \left(\frac{x_0 - \sqrt{a}}{x_0 + \sqrt{a}}\right)^{2^n}$, deduce quadratic convergence with an explicit constant, and identify the substitution of Theorem 1 d) which makes all of this one line.